Multiple Linear Regression

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Model

$$y = \alpha + \beta_1 \cdot x_1 + \cdots + \beta_n \cdot x_n + \varepsilon$$

- y: Dependent variable
- x_i: Independent variables
- α: y-intercept
- β_i: Slope coefficients
- ε: Unknown, true population errors

Model

$$y = \alpha + \beta_1 \cdot x_1 + \cdots + \beta_n \cdot x_n + \varepsilon$$

- We seek to explain y with x_i's
- "Regress y onto x"
- Observe a sample of y and x
- Estimate α and β_i
  - Estimates denoted $\hat{\alpha}$ and $\hat{\beta}_i$
- Do not observe ε
  - Assume $\varepsilon \sim N(0, \sigma^2)$
  - σ^2 unknown and needs to be estimated

Methodology

1. Estimate model
2. Inference: Is the model is good statistically?
   - ANOVA
   - Hypothesis Tests
3. Prediction: Use the model to explain and predict.

Model

Unobserved, true population model

$$y = \alpha + \beta_1 \cdot x_1 + \cdots \beta_n \cdot x_n + \varepsilon$$

- α, β_i unobserved true parameters
- ε unobserved population errors
- Observe subsample y and x

Model Estimation

Estimate the model

$$\hat{y} = \hat{\alpha} + \hat{\beta}_1 \cdot x_1 + \cdots + \hat{\beta}_n \cdot x_n$$

- $\hat{\alpha}$, $\hat{\beta}_i$ are estimated
- $\hat{y}$ is the predicted value given x and estimated $\hat{\alpha}$ and $\hat{\beta}_i$

Sample Error

Sample Error

$$e = y - \hat{y}$$

- e is the sample error from the estimation
- $y = \hat{\alpha} + \hat{\beta} \cdot x + e$

Estimation

- Goal is to choose $\hat{\alpha}, \hat{\beta}$ to find the "best" line
- How do we define best? Minimize the sum of the squared errors

$$\min_{\{\alpha,\beta\}} \sum_{i=1}^{n} e_i^2$$

Regression Application

Estimate the model

$$EXR_i = \hat{\alpha}_i + \hat{\beta}_1 \cdot F_1 + \cdots + \hat{\beta}_n \cdot F_n + e_i$$

- $\hat{\alpha}_i$ has the interpretation of a pricing error. If the correct factor pricing model was $\{F_1, \ldots, F_n\}$, then the factors should capture and explain asset i's risk premium and $\alpha_i = 0$

- Technical Note: For α_i to truly be a pricing error, the following must be true
  - EXR_i is the excess return of the asset ($R_i - R_f$)
  - Each F_i are differences in returns, e.g. $R_{Small} - R_{Big}$

Sum of Squares

$$SS = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- Measure of variation
- $\bar{x}$ is the average of x
- Excel function: $SS = \mathrm{devsq}(x)$

Sum of Squares

SS	Name	Variable	Formula	Interpretation
SSX	Sum of squares of independent variable	x	$\sum_{i=1}^{n}(x_i - \bar{x})^2$	Independent variable
SST	Total sum of squares	y	$\sum_{i=1}^{n}(y_i - \bar{y})^2$	Dependent variable
SSE	Sum of squared errors	e	$\sum_{i=1}^{n}(e_i - \bar{e})^2$	Error
SSR	Sum of squares of regression	$\hat{y}$	$\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$	Regression Model

ANOVA

Decompose variation in data y, i.e. the SST, into the sum of:

- variation explained by the regression model, SSR
- variation unexplained by model, i.e. the error SSE

$$y = \hat{y} + e$$
$$SS(y) = SS(\hat{y}) + SS(e)$$
$$SST = SSR + SSE$$
$$\text{Data} = \text{Model} + \text{Error}$$

ANOVA

	Degrees of Freedom $(\mathbf{df})$	**Sum of Squares** $(\mathbf{SS})$	**Mean Squared** $\left(\mathbf{MS} = \frac{\mathbf{SS}}{\mathbf{df}}\right)$	**F-stat**
Regression (Model)	$df_R = k = \#\beta s$	SSR	$MSR = \frac{SSR}{df_R}$	$F = \frac{MSR}{MSE}$
Error	$+df_e = n - (k+1)$	$+SSE$	$MSE = \frac{SSE}{df_e}$	
Total (Data)	$= df_T = n - 1$	$= SST$		

Adjusted $\bar{R}^2$

- R^2 mechanically always increases as you add more x_i's
- Cannot compare R^2's from models with different number of x_i's
- Instead, we use adjusted $\bar{R}^2$

$$\bar{R}^2 = 1 - \frac{SSE}{SST} \times \frac{df_T}{df_e}$$

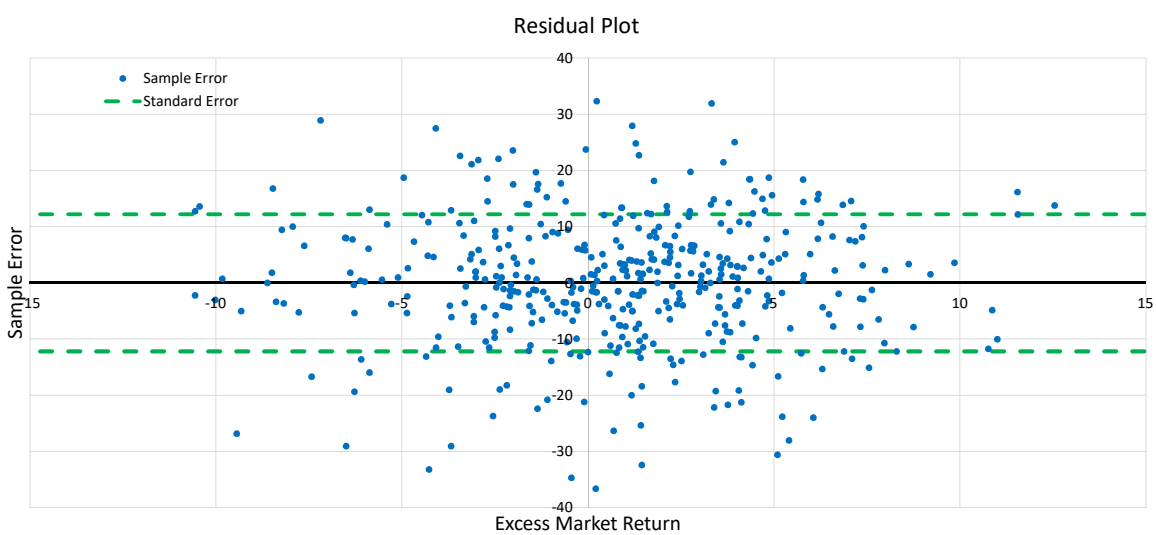
- Recall that
  - $SST = SSR + SSE$
  - $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$
- $\frac{df_T}{df_e}$ is penalty for more x_i's: $k \uparrow \Rightarrow \bar{R}^2 \downarrow$

Standard Error of Regression

- s is the standard error of the regression
  - The standard error of the regression is approximately the standard deviation of the errors (but we can't use the stdev() formula): $s \approx SD(e)$

$$s = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}(e_i - \bar{e})^2} = \sqrt{MSE}$$

Standard Error of Regression



Hypothesis Tests

- Now that we can estimate numerous models, we want to know which models are best statistically
- When we perform a hypothesis test, we will always compare two hypothesis: a null and an alternative
- Goal is to see which hypothesis is supported by data statistically

Hypothesis Tests: F-test

- Null hypothesis:

$$H_0: \beta_1 = \cdots = \beta_k = 0$$

- Alternative hypothesis:

$$H_a: \text{At least one of } \beta \neq \beta_0$$

Hypothesis Tests: F-test

- Two methods to conduct hypothesis test
  - Both methods will give identical conclusion
  1. Compare x-axis: F-stat vs critical value
  2. Compare y-axis: P-value vs significance level

F-distribution

- F-distribution is related to the square of the t-distribution
- It is bounded from 0 to ∞
- Extra parameters: degrees of freedom
  - For hypothesis tests, use r and $n-(k+1)$ as the degrees of freedom for the numerator and denominator
- Excel functions f.dist and f.inv similar to norm.dist and norm.inv
- Probability of $0 \leq t \leq a$ is the area under curve from 0 to a

$$P(0 \leq t \leq a) = \mathrm{f.dist}(t, df_n, df_d, \mathrm{TRUE})$$

- The value on the x-axis, a, that corresponds to probability $p = P(0 \leq t \leq a)$

$$a = \mathrm{f.inv}(p, df_n, df_d)$$

Subjective calculations

- Significance level α
  - y-axis
  - Probability of incorrectly rejecting null
  - Typical levels of 1%, 5%, or 10% in practice
  - If you want to be more confident of result, use smaller α
    - Confidence $= (1-\alpha)100\%$
    - Smaller α is a tougher hurdle to discover a result

- Critical value
  - x-axis

$$\alpha = \mathrm{Prob}(x \geq cv) \Rightarrow \mathrm{cv} = \mathrm{f.inv}(1-\alpha, df_n, df_d)$$

Methods 1 & 2

Method 1: t-stat vs critical value

- Reject null if $\mathrm{F-stat} \geq cv$

Method 2: P-value vs significance level

- Reject null if P-value $<$ significance level

F-tests

- Model comparison

$$\begin{aligned} \text{Full: } y &= \alpha + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \beta_3 \cdot x_3 + \varepsilon \\ \text{Reduce: } y &= \alpha + \beta_1 \cdot x_1 + \varepsilon \end{aligned}$$

- Hypothesis

$$\begin{aligned} &H_0: \beta_2 = \beta_3 = 0 \\ &H_1: \text{At least one of } \{\beta_2, \beta_3\} \neq 0 \end{aligned}$$

- Conclusion
  - $H_0 \Rightarrow$ Reduce model is better statistically
  - $H_1 \Rightarrow$ Full model is better statistically

F-tests

- n number of observations
- k number of β's in full model
- r number of β's testing if can drop

$$F - \text{stat} = \frac{SSE_R - SSE_F}{r \cdot MSE_F}$$

- P-value
  - y-axis
  - "Probability of estimating $\hat{\beta}$ given the null is true"

$$\text{P} - \text{val} = \text{Prob}(x \geq F - stat) = 1 - \text{f.dist}(F - stat, df_n, df_d)$$

F-tests & F-distribution

- F-distribution is truncated at 0 on the left
- Always perform a one-sided test to the right
- Degrees of freedom r and $n - (k + 1)$

Model Selection

- Forward Selection
  - Start with no factors and determine the most important factors to add one at a time
  - Stop when there are no more important factors to add
- Backward Selection
  - Start with all factors and determine the least important factors to drop one at a time
  - Stop when there are no more unimportant factors to drop
- Stepwise Regression
  - Perform one backward selection and then one forward selection
  - Continue alternating until there are no more factors to add or drop

Prediction

- Predict y with x:

$$\hat{y} = \hat{\alpha} + \hat{\beta}_1 \cdot x_1 + \cdots + \hat{\beta}_n \cdot x_n$$

Qualitative (Categorical) Independent Variables

- Quantitative variables: Variables taking numeric values
  - Example: GPA $x = 3.7$
- Qualitative variables: Variables taking categories
  - Example: Class $x = \text{Junior}$

- Use indicator (dummy) variables to convert qualitative variables into quantitative variables

$$x = \begin{cases} 0 & \text{if Junior} \\ 1 & \text{if Senior} \end{cases}$$

$$y = \beta_1 x_1 + \beta_2 x_2$$

Grade GPA Sex

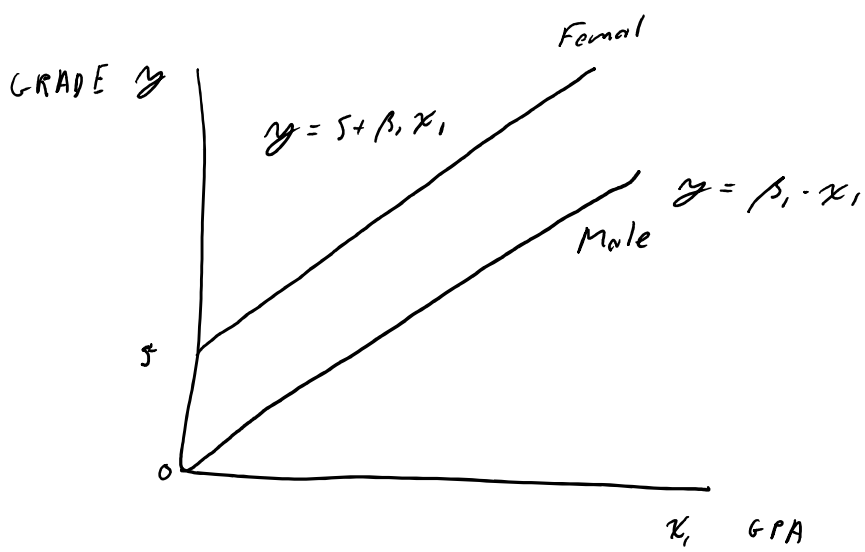
Shift intercept

$\beta_1 = 10$ ↑ GPA 1 pt => Grade ↑ 10 pt

$\beta_2 = 5$ Females have 5 pt higher grade

Male $\quad$ Grade $= 10 \cdot GPA + 5 \cdot 0 = 10 \cdot GPA$

Female $\quad$ Grade $= 10 \cdot GPA + 5 \cdot 1 = 5 + 10 \cdot GPA$



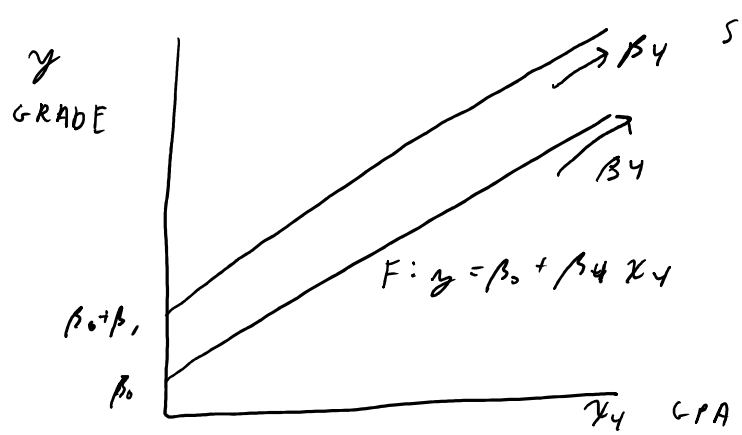
Multiple : Class

Have 1 less dummies than categ.

	x_1	x_2	x_3
Freshman	0	0	0
Soph	1	0	0
Jun	0	1	0
Sen	0	0	1

$$y = \beta_0 + \underbrace{\beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3}_{\text{Class}} + \beta_4 x_4$$

Grade GPA



$S: y = (\beta_0 + \beta_1) + \beta_4 x_4$

Interaction

x_1 GPA

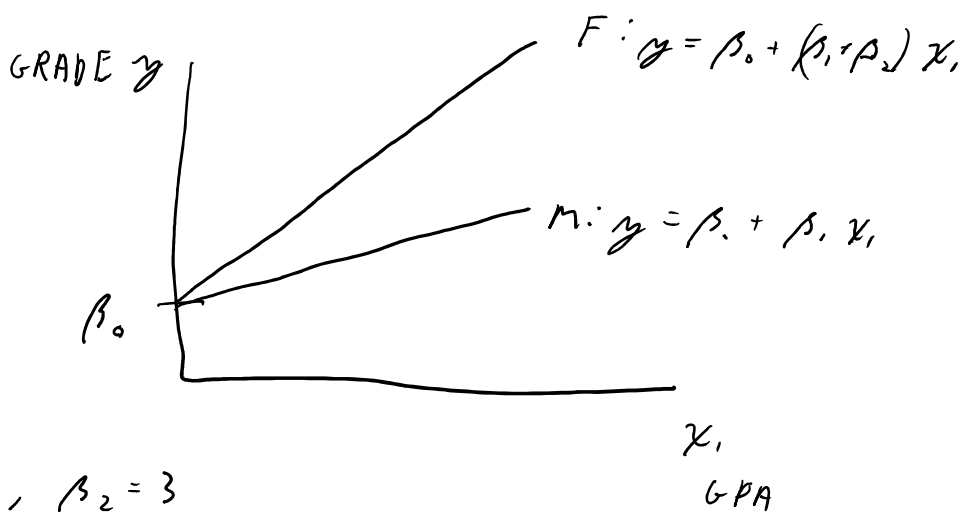
$x_2 = \begin{cases} 0 & \text{Male} \\ 1 & \text{Female} \end{cases}$ Sex

Affects Slopes

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_1 \cdot x_2$$

F: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_1 \cdot 1 = \beta_0 + (\beta_1 + \beta_2) x_1$

M: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_1 \cdot 0 = \beta_0 + \beta_1 x_1$



$\beta_1 = 5$, $\beta_2 = 3$

$\Rightarrow$ F: $\uparrow$ GPA 1 pt $\Rightarrow$ $\uparrow$ GRADE 8 pt

M: " " 5 pt

Non-Linear Models

To capture non-linear relationships, can regress on

- Higher order interactions: $x^2, x^3, \ldots$

$$y = \alpha + \beta_1 \cdot x + \beta_2 \cdot x^2 + \varepsilon$$

- Non-linear transformations: $ln(x)$

$$y = \alpha + \beta_1 \cdot ln(x) + \varepsilon$$